

FOgym: A Bottom-Up Home Energy Management Framework Based on FlexOffers and Multi-Agent Reinforcement Learning

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Abstract

The FlexOffer (FO) has been widely adopted as a standardized data model for representing flexible electricity consumption and generation, particularly for modern energy management systems that require both adaptability and flexibility. However, existing FO generation methods rely on predefined models, which limit their adaptability in dynamic and complex environments. To address this limitation, we propose FOgym, the first unified bottom-up energy management framework that integrates multi-agent reinforcement learning (MARL) into the FO life-cycle, eliminating the need for explicit modeling. FOgym introduces a model-free FO generation approach based on MARL, where agents learn optimal behaviors through real-time interaction with the environment. This enables the system to adapt flexibly to uncertainty without depending on prior knowledge or hand-crafted models. To integrate MARL uniformly with the FO life-cycle pipeline, we design cross-stage observation spaces that incorporate downstream signals and closed-loop rewards that propagate pipeline outcomes, enabling end-to-end optimization. The resulting FOgym environment features modular, customizable algorithmic components for generation, aggregation, trading, and scheduling, supporting evaluation in bottom-up paradigms. Extensive experiments conducted in diverse environments demonstrate that FOgym not only enables effective FO generation and management using MARL, but also provides a unified framework for benchmarking various methods, consistently outperforming state-of-the-art approaches in efficiency and adaptability, while different algorithms exhibit distinct strengths suited to various deployment scenarios. Code is available at <https://anonymous.4open.science/r/FOgym-3650/>.

CCS Concepts

• **Computing methodologies** → **Reinforcement learning**; **Multi-agent systems**; • **Applied computing** → **Smart grid**.

Keywords

FlexOffer, Multi-Agent Reinforcement Learning, Home Energy Management, Bottom-Up Coordination, Smart Grid

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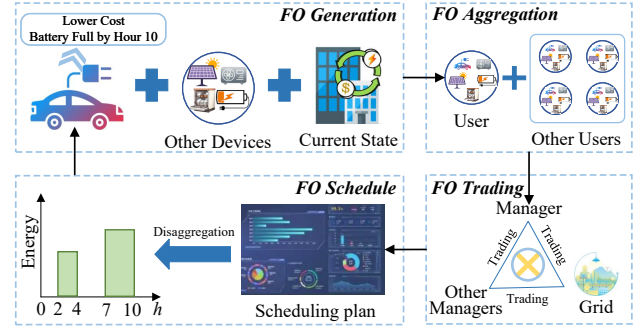


Figure 1: Example of EV for FO life-cycle

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1 Introduction

Decarbonization is turning centralized power systems into complex distributed energy resource (DER) networks, posing new coordination challenges under uncertainty [38, 40]. Existing energy management systems include three architectures: top-down centralized control, flat distributed control, and bottom-up decentralized coordination [18]. Top-down centralized systems offer theoretical optimality but suffer from scalability limitations, communication bottlenecks, and privacy concerns when applied to large-scale distributed systems [20, 37]. In response, flat distributed architectures have emerged to support decentralized control [28], and their local decision-making structure aligns well with multi-agent reinforcement learning (MARL). However, flat architectures lack the structural capacity for global coordination and forward-looking decision-making, which limits their effectiveness in dynamic market environments and often results in locally optimal but system-inefficient outcomes. To address these limitations, bottom-up architectures enable a structured transition from device-level flexibility to higher-level managerial coordination. However, the complexity of coordination, uncertainty, and operational constraints within bottom-up architectures make it difficult to achieve system-level optimality.

The FlexOffer (FO) model, originating from major European research projects [4, 23, 31], addresses this bottleneck by providing a device-independent, standardized, scalable, and high-accuracy approximation of flexibility. The FO life-cycle—covering generation, aggregation, trading, and scheduling—offers a complete workflow for managing large-scale distributed assets [25].

As a concrete example, consider an EV user who connects their vehicle to a home charging station after work and departs the next morning, as illustrated in Fig. 1. Within this time window, the system generates an FO representing the EV's charging flexibility, which is then aggregated with other FOs by a regional manager

for market participation. After trading, successful FOs are disaggregated and scheduled back to individual users who bought them. This, in turn, enables the EV to charge at optimal times while minimizing energy costs as well.

Despite these advantages, current FO research remains predominantly model-driven [13, 14, 27, 29–31]. The lack of a unified end-to-end pipeline that supports integrated optimization across stages limits reproducibility, coordination, and practical large-scale deployment. To overcome these limitations, we have two key challenges.

Challenge I: How to equip the FO system with real-time adaptation and autonomous optimization in dynamic, uncertain environments? The current studies on FO generation primarily focus on building upon formal models that describe energy flexibility using constraints, variables, and optimization objectives. These studies are predominantly model-based, relying on explicit modeling of flexibility and employing algorithmic or optimization techniques based on these models [13–15]. While such approaches can yield near-optimal results under ideal assumptions, their performance is highly dependent on model fidelity [22, 42]. Modeling the stochastic and nonlinear behavior of DERs, such as EVs or renewables, is inherently difficult. Moreover, static models fail to adapt to time-varying dynamics like equipment aging or shifting user preferences.

Challenge II: How to integrate MARL uniformly across the multi-stage FO life-cycle? The FO life-cycle spans multiple tightly coupled stages, including generation, aggregation, trading, and scheduling, where upstream decisions directly shape downstream outcomes. For instance, an FO with overly aggressive flexibility bounds may fail during aggregation due to constraint violations [27], or succeed in aggregation but receive unfavorable prices in trading due to market mismatch [31]. However, existing MARL approaches for energy management typically optimize each stage in isolation. General-purpose algorithms such as multi-agent deep deterministic policy gradient (MADDPG) [17] and Q-mixing network (QMIX) [39] provide foundational multi-agent coordination but lack pipeline-aware mechanisms. When applied to energy systems, hierarchical methods [19, 24] coordinate agents across spatial scales, and market-based approaches [2] focus on trading optimization, yet none address the temporal dependencies across the FO life-cycle. Consequently, generation agents cannot perceive whether their FOs lead to successful trades or profitable outcomes, preventing end-to-end optimization.

To tackle these challenges, we propose FOGym, a novel bottom-up home energy management framework based on FO, explicitly designed to integrate MARL methods into the complete FO life-cycle. To address Challenge I, FOGym adopts MARL as a model-free method for FO generation, enabling agents to learn optimal strategies through real-time interaction with the environment without requiring explicit device modeling. This allows the system to adapt dynamically to uncertain conditions such as user behavior changes and market fluctuations. To address Challenge II, FOGym designs pipeline-aware observation spaces that incorporate downstream signals (e.g., market prices, trading outcomes) and closed-loop reward functions that propagate pipeline outcomes back to the generation stage. This feedback-aware formulation enables generation agents to perceive downstream outcomes and learn to generate

market-aware FOs, achieving end-to-end optimization across the FO life-cycle. Our main contributions are as follows.

- We present FOGym, a novel bottom-up home energy management framework based on FOs to enable standardized benchmarking and integration of MARL across the complete FO life-cycle.
- FOGym introduces a model-free FO generation method based on MARL, which eliminates the need for explicit device modeling and enables real-time adaptation to dynamic and uncertain environments.
- FOGym integrates MARL uniformly with the FO life-cycle pipeline through cross-stage observation and closed-loop reward design, enabling generation agents to perceive downstream outcomes and achieve end-to-end optimization.
- Experiments conducted in a real-world energy system demonstrate that FOGym enables effective MARL-based FO management and supports comprehensive evaluation of diverse methods within a unified bottom-up framework.

2 Related Work

2.1 FO-Based Energy Management

FO research has evolved from standardized flexibility modeling to a complete life-cycle covering generation, aggregation, trading, and disaggregation. Foundational work [25] established the FO definition and life-cycle framework, demonstrating scalable management of millions of FOs with high accuracy in smart grid environments. Building on this foundation, subsequent research extended FO generation to diverse device types: a general and scalable method was proposed for capturing battery flexibility [13], while Heat FOs [14] introduced device-independent representations for electricity-heat coupling in thermal systems. Systematic methods were developed for generating and evaluating FOs from flexible electrical devices including EVs and wet appliances [22]. To address real-world uncertainty, Uncertain FOs [15] incorporated probabilistic models to capture stochastic behavior in energy flexibility. The GOFLEX project [23] validated FO-based coordination at scale, deploying the complete FO life-cycle across three European cities with over 500 prosumers. Despite these advances, existing FO research remains predominantly model-based, and there is a lack of unified environments that support learning-based approaches for FO generation and management.

2.2 MARL for Energy Systems

Reinforcement learning has emerged as a promising solution for distributed coordination in complex energy systems [12, 21]. Single-agent methods, including value-based approaches like Deep Q-networks [34], policy gradient methods such as Proximal Policy Optimization (PPO) [8] and Trust Region Policy Optimization [35], and actor-critic algorithms including Deep Deterministic Policy Gradient (DDPG) [41] and Soft Actor-critic [7], have been effective for individual device-level control tasks such as battery scheduling and demand response. However, these methods inherently lack scalability to handle multi-stakeholder interactions in distributed energy systems where multiple agents must coordinate their decisions. MARL methods address these limitations by enabling coordinated learning among multiple agents [10]. These methods are typically categorized by their training and execution paradigms

[5]: centralized training with centralized execution (CTCE) methods such as Deep Multi-agent Parameterized Q-network [6] and JointPPO [16] require full observability during both training and deployment; centralized training with decentralized execution (CTDE) methods including QMIX [39] and MADDPG [17] learn coordinated policies with centralized critics while enabling decentralized deployment; and decentralized training with decentralized execution (DTDE) methods like Flexible FULLY-decentralized Approximate Actor-critic [11] and Momentum-based Decentralized Policy Gradient Tracking [9] achieve fully decentralized learning. While these MARL methods have been applied to energy trading and demand response, they typically optimize single-stage objectives without considering cross-stage dependencies in multi-stage energy management workflows such as the FO life-cycle.

3 Preliminaries

3.1 FlexOffer

An FO f is defined as a tuple $f = ([t_{es}, t_{ls}], p)$, where $[t_{es}, t_{ls}]$ represents the start time flexibility interval and p is the energy amount profile. The time flexibility is characterized by the earliest start time t_{es} and the latest start time t_{ls} , allowing the flexible resource to adjust when its operation begins. The energy profile p consists of a sequence of time slices $\langle s^1, s^2, \dots, s^n \rangle$, where each slice s^i is defined by a continuous range $[e_{min}^i, e_{max}^i]$ specifying the energy bounds for that time interval. Additionally, the FO includes a Total Energy Constraint (TEC) $e_{min} \leq \sum_{i=1}^n e^i \leq e_{max}$, where $e^i \in [e_{min}^i, e_{max}^i]$ is the actual energy in slice i , and e_{min} and e_{max} represent the minimum and maximum total energy across all slices [25]. A visual representation of a simple FO is shown in Fig. 2.

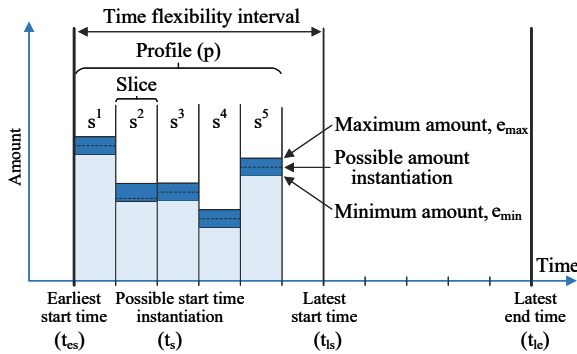


Figure 2: A visual representation of a simple FO

3.2 Dec-POMDP

The problem of MARL for cooperation among agents is conventionally formulated as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP) [26]. Formally, a Dec-POMDP is defined as a tuple $M_{Dec} = \langle \mathcal{D}, \mathcal{S}, \mathcal{A}, T, O, \mathcal{O}, R, h, b_0 \rangle$, where $\mathcal{D} = \{1, 2, \dots, n\}$ is the set of n agents; $\mathcal{S}$ is a finite set of world states; $\mathcal{A} = \times_{i \in \mathcal{D}} \mathcal{A}_i$ is the set of joint actions, where $\mathcal{A}_i$ is the action set for agent i ; $T : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ is the state transition function specifying $\Pr(s'|s, a)$; $\mathcal{O} = \times_{i \in \mathcal{D}} \mathcal{O}_i$ is the set of joint observations, where $\mathcal{O}_i$ is the observation set for agent i ; $O : \mathcal{A} \times \mathcal{S} \times \mathcal{O} \rightarrow [0, 1]$ is the observation function specifying $\Pr(o|a, s')$; $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$

is the immediate reward function shared by all agents; h is the planning horizon; $b_0 \in \Delta(\mathcal{S})$ is the initial state distribution. In the following sections, we formulate the FO life-cycle as a Dec-POMDP, where managers act as agents that generate FOs based on partial observations of device states and market conditions.

4 Problem Statement and Framework

This section presents the FO life-cycle framework and defines the objective of FOgym. We first describe the bottom-up FO life-cycle with formal definitions for each stage, then state the optimization objective that FOgym aims to achieve. A concrete example illustrating how an EV participates in the complete FO life-cycle is provided in Appendix A.1.

4.1 Bottom-up FO Life-cycle

The FO life-cycle represents a complete bottom-up framework for managing controllable distributed energy flexibility, spanning from device-level resource identification to market-based trading and real-time scheduling execution, as shown in Fig. 3. It consists of four sequential modules, namely generation, aggregation, trading, and scheduling, with the overarching goal of minimizing energy waste, reducing user costs, and enhancing demand-side responsiveness and user satisfaction. The definition of each module is as follows:

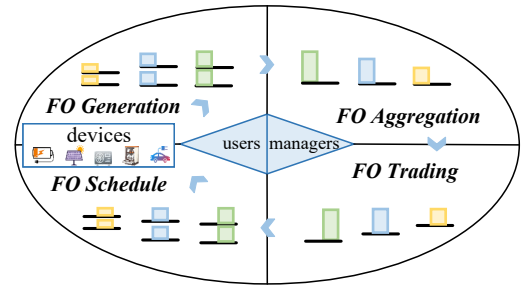


Figure 3: The FO life-cycle process view

Generation. The first stage extracts flexibility from individual devices and represents it as FOs.

DEFINITION 1. A generated FO from device $d \in \mathcal{D}$ at time t is defined as $F_d(t) = \{f_{s,d,i}(t)\}$, where $f_{s,d,i}(t) = ([t_s^i, t_e^i], [e_{min}^i, e_{max}^i], p_d, c_d)$ represents a FO slice with start time t_s^i , end time t_e^i , energy bounds $[e_{min}^i, e_{max}^i]$ (in kWh), device-specific parameters p_d (e.g., charging power, efficiency), and operational constraints c_d (e.g., SOC bounds, temperature limits), while $i \in [1, n_s]$ and n_s is the number of FO slices.

While the standard FO definition in Section 3.1 captures start time flexibility at the FO level using $[t_{es}, t_{ls}]$, the generated FO assigns each time slice concrete time boundaries $[t_s^i, t_e^i]$ within the flexibility interval, representing the actual execution window for that slice. For example, an EV arriving at 18:00 with a 60 kWh battery at 20% state-of-charge (SOC), requiring 80% SOC by 7:00 next morning, generates an FO with $t_s = 18 : 00$, $t_e = 7 : 00$, $e_{min} = 36$ kWh (to reach 80% SOC), and $e_{max} = 48$ kWh (to reach full charge).

Aggregation. The second stage combines individual FOs from multiple devices under a manager into aggregated FOs.

DEFINITION 2. An aggregated FO from manager $m \in \mathcal{M}$ at time $t \in T$ is defined as $F_m^A(t) = ([t_s^m, t_e^m](t), [e_{min}^m, e_{max}^m](t), P_m(t))$, where $[t_s^m, t_e^m]$ represents the time window, $[e_{min}^m, e_{max}^m]$ represents the aggregated energy bounds with $e_{min}^m = \sum_{d \in \mathcal{D}_m} e_{min,d}$ and $e_{max}^m = \sum_{d \in \mathcal{D}_m} e_{max,d}$, and P_m encodes the flexibility profile.

Trading. The third stage enables managers to trade energy flexibility in the market based on their aggregated FOs.

DEFINITION 3. A set of bids formulated by manager $m \in \mathcal{M}$ based on its aggregated FO is defined as $B_m(t) = (E_m^b(t), p_m^b(t), k_m(t))$, where $E_m^b(t) \in [e_{min}^m, e_{max}^m](t)$ represents the energy quantity offered for trade, $p_m^b(t) = p_b^t \times (1 \pm \phi_a \pm \psi_r \pm b_m)$ represents the bid price with base price adjustment p_b^t , market dynamics factor ϕ_a , controlled randomness $\psi_r \in [-1.5, 1.5]\%$, and manager-specific bias b_m , and $k_m(t) = \{0, 1\}$ indicates the bid direction, 0 indicates selling, and 1 indicates buying.

For example, if a manager wants to sell 50 kWh of surplus flexibility and the current market price is 0.3 DKK/kWh, the manager may submit a sell bid at $p_m^b = 0.3 \times 0.98 = 0.294$ DKK/kWh to increase the chance of a successful trade.

DEFINITION 4. A set of successful bilateral trades at time $t \in T$ is defined as $\mathcal{T}(t) = M_c(B_m(F_m^A(t)))$, where M_c represents the market trading mechanism.

Scheduling. The final stage disaggregates successful trades back to individual devices as control signals.

DEFINITION 5. For a manager who successfully purchased energy through trading, a device-specific control signal after disaggregation is defined as $C_d(t) = (P_d^c(t), t_d^s, t_d^e)$, where $P_d^c \in [p_{min,d}, p_{max,d}]$ represents the power command for device d , and t_d^s and t_d^e represent the execution time windows, subject to device constraints and user preferences.

4.2 Objective of FOgym

FOgym aims to provide a model-free solution for FO generation that matches the key objectives traditionally achieved by model-based approaches, including minimizing overall energy waste, reducing user energy costs, and satisfying user energy demands. Instead of relying on explicitly defined system models, FOgym leverages MARL to learn FO generation policies through continuous interaction with the environment. This enables the system to adapt online to changing conditions such as user behavior, device availability, and market signals. Furthermore, FOgym aims to integrate MARL uniformly across the multi-stage FO life-cycle by incorporating downstream signals into observations and propagating trading outcomes through rewards. This enables generation agents to perceive how their FOs perform in subsequent stages and learn to generate market-aware FOs, achieving end-to-end optimization across the entire life-cycle.

5 FOgym

Fig. 4 shows the FOgym framework that includes four modules: generation, aggregation, trading, and scheduling. We first formulate the FO generation problem as a Dec-POMDP, enabling model-free FO generation via MARL without explicit device modeling. We then

present a cross-stage feedback mechanism that incorporates downstream signals into observation spaces and propagates pipeline results through reward functions, enabling end-to-end optimization across the FO life-cycle. Finally, we describe the modular life-cycle implementation with standardized interfaces and evaluation metrics.

5.1 MARL-based Model-free FO Generation

To enable real-time adaptation in dynamic, uncertain environments without explicit device modeling, we adopt the MARL paradigm and reformulate the FO generation problem as a Dec-POMDP. This formulation allows agents to learn optimal FO generation strategies through direct interaction with the environment, eliminating the need for accurate system models that are difficult to obtain for stochastic DERs. Agents iteratively update policies via decentralized interaction and experience accumulation, achieving adaptive FO generation that automatically adjusts to time-varying dynamics.

Dec-POMDP MARL Problem Formulation. We formulate the FO coordination problem as a Dec-POMDP:

$$\langle \mathcal{M}, \mathcal{S}, \{\mathcal{A}_m\}, \{\mathcal{O}_m\}, T, R, \gamma \rangle, \quad (1)$$

where $\mathcal{M} = \{1, 2, \dots, M\}$ denotes the set of M manager agents (corresponding to $\mathcal{D}$ in the standard Dec-POMDP formulation, with managers acting as decision-making agents); $\mathcal{S}$ represents global state space (unobservable to individual agents); $\mathcal{A}_m \subseteq \mathbb{R}^{d_m}$ represents the continuous action space of agent m ; $\mathcal{O}_m \subseteq \mathbb{R}^{o_m}$ represents observation space of agent m ; $T : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ represents state transition function; $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ represents the system reward function shared by all agents; $\gamma \in [0, 1]$ represents discount factor.

Unified Action Space for FO Generation. To handle device heterogeneity, FOgym defines a unified action representation that accommodates diverse operational constraints while preserving device-specific characteristics. Each manager's action $a_m(t)$ generates FO parameters for all devices under its control:

$$a_m(t) = \{(f_d^{ts}, f_d^{te}, e_d^{min}, e_d^{max}, w_d) \mid d \in \mathcal{D}_m\}, \quad (2)$$

where $f_d^{ts}, f_d^{te} \in [-1, 1]$ represent start and end time flexibility factors that are scaled to the device's available time window (e.g., -1 shifts to the earliest feasible time, 1 to the latest); $e_d^{min} \in [0.1, 1.0]$ and $e_d^{max} \in [1.0, 2.0]$ represent minimum and maximum energy factors; $w_d \in [0.1, 2.0]$ represents priority weight. This unified interface abstracts heterogeneous devices (EVs, batteries, heat pumps) into a common FO parameter space while device-specific constraints are enforced through post-processing.

MARL Algorithm Suite. To solve the Dec-POMDP formulation, FOgym integrates five representative MARL algorithms spanning different coordination paradigms. Each algorithm is adapted with FO-specific modifications to ensure compliance with the FO pipeline.

- **MAPPO** employs shared policy networks with trust-region constraints, providing stable training through clipped surrogate objectives and generalized advantage estimation. The shared architecture enables efficient parameter sharing across managers, suitable for homogeneous configurations. For FO adaptation, the loss function is augmented as:

$$\mathcal{L}^\pi = \mathcal{L}_p + \lambda_v \mathcal{L}_v - \lambda_e \mathcal{H} + \lambda_f \mathcal{L}_f + \lambda_t \mathcal{L}_t, \quad (3)$$

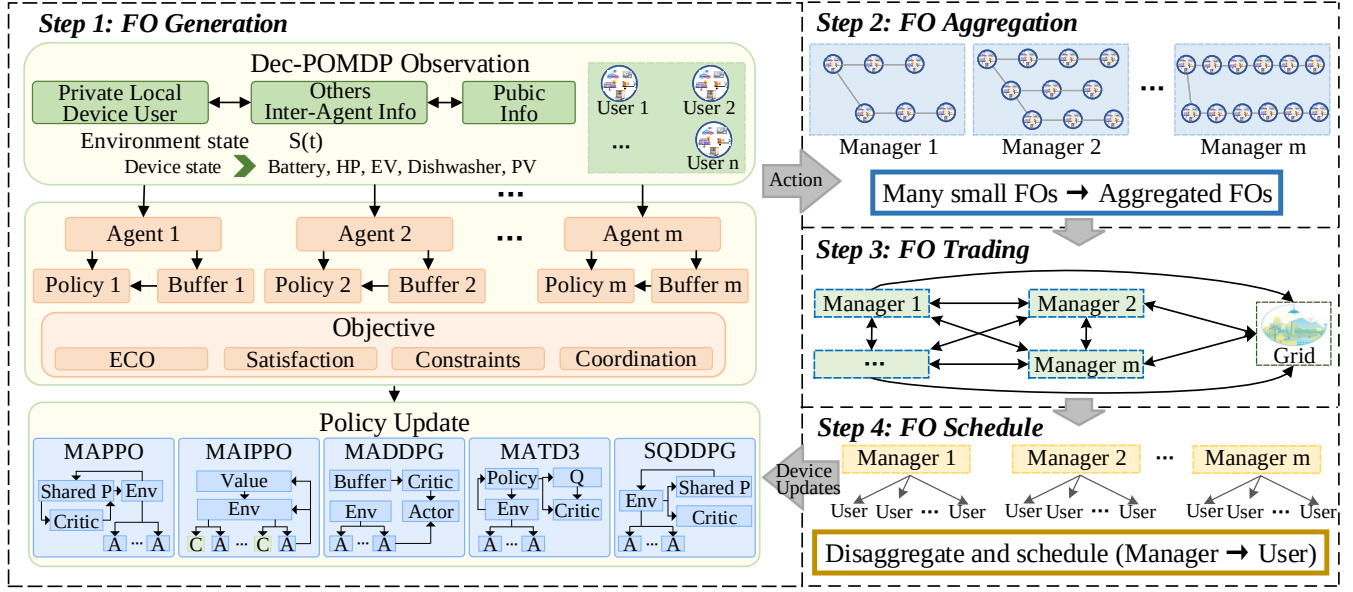


Figure 4: Overview of the FOgym framework

where $\mathcal{L}_p$ is the clipped surrogate objective; $\mathcal{L}_v$ is the value function loss; $\mathcal{H}$ is the entropy bonus for exploration; $\mathcal{L}_f = \sum_d \max(0, |a_d^t - \bar{a}_d|) + \max(0, |a_d^e - \bar{a}_d|)$ penalizes violations of FO temporal and energy constraints; $\mathcal{L}_t = -\sum_m s_m^p \cdot \eta_m^k$ incorporates market feedback by weighting trading success s_m^p and bidding efficiency η_m^k ; $\lambda_v, \lambda_e, \lambda_f, \lambda_t$ are the corresponding weighting coefficients.

- *Multi-Agent Independent PPO (MAIPPO)* assigns independent actor-critic networks to each manager, enabling personalized policy adaptation for heterogeneous user-device portfolios. This architecture prevents negative transfer between managers with different device compositions. Each manager independently optimizes:

$$\pi_m^* = \arg \max_{\pi_m} \mathbb{E}[\mathcal{L}_p(\pi_m) + \lambda_f^m \mathcal{L}_f^m], \quad (4)$$

where π_m is the policy of manager m ; $\mathcal{L}_p(\pi_m)$ is the standard PPO objective for manager m ; $\mathcal{L}_f^m$ is the FO constraint loss specific to manager m 's device portfolio; λ_f^m is the manager-specific weighting coefficient.

- *MADDPG* utilizes centralized critics with decentralized actors, achieving sample-efficient learning through deterministic policy gradients and experience replay. The CTDE paradigm allows critics to access global information during training. FO adaptation incorporates constraint regularization into the critic loss:

$$\mathcal{L}_c = \mathcal{L}_q + \lambda_c \sum_d \max(0, |a_d| - \bar{a}_d)^2, \quad (5)$$

where $\mathcal{L}_q$ is the temporal difference loss for Q-value estimation; a_d is the action for device d ; $\bar{a}_d$ is the operational bound for device d ; λ_c is the constraint penalty coefficient. This regularization penalizes actions that violate device operational bounds.

- *Multi-Agent Twin Delayed DDPG (MATD3)* extends MADDPG with twin Q-networks and delayed policy updates, mitigating value overestimation bias for improved robustness. The FO-specific

adaptation uses the minimum of two critics:

$$Q^T(s, a) = \min(Q_1^T(s, a), Q_2^T(s, a)), \quad (6)$$

where Q_1^T and Q_2^T are two independent critic networks with FO-aware value estimation; taking the minimum reduces overestimation bias. Target policy smoothing is applied as $a' = \pi'(s') + \text{clip}(\epsilon, -c, c)$, where $\epsilon \sim \mathcal{N}(0, \sigma)$ is exploration noise clipped within $[-c, c]$ to reduce variance in volatile market conditions.

- *Shapley Q-value DDPG (SQDDPG)* incorporates Shapley-value-based credit assignment, computing each manager's marginal contribution to the collective reward. The fairness-aware critic loss is:

$$\mathcal{L}^S = \mathcal{L}_q + \lambda_s |\phi_m - \hat{\phi}_m|^2, \quad (7)$$

where $\mathcal{L}_q$ is the temporal difference loss; ϕ_m is the computed Shapley value representing manager m 's marginal contribution to the collective reward; $\hat{\phi}_m$ is the target Shapley value; λ_s is the fairness weighting coefficient. This mechanism ensures equitable reward distribution among cooperative agents.

The details of each algorithm for the purpose of integration are provided in Appendix B.1.

5.2 Cross-stage Feedback Mechanism

Traditional MARL approaches for energy management optimize only the generation stage in isolation, treating downstream modules (aggregation, trading, scheduling) as black boxes. This creates a fundamental limitation: agents cannot learn how their FO generation decisions affect market outcomes or user satisfaction. For instance, an agent might generate FOs with high flexibility but poor market compatibility, leading to failed trades.

To address this limitation, FOgym designs a feedback-aware formulation that enables agents to perceive downstream outcomes through two complementary mechanisms: (i) a *cross-stage observation structure* that incorporates downstream signals (market prices,

inter-manager states) into the observation space, enabling agents to anticipate how their FOs will perform in subsequent stages; and (ii) a *closed-loop reward design* that propagates trading success, economic efficiency, and user satisfaction back to the generation stage. Together, these mechanisms achieve end-to-end optimization across the entire FO life-cycle while preserving modular flexibility. **Cross-stage Observation Structure.** To enable agents to perceive downstream pipeline signals and adapt their FO generation strategies accordingly, each manager's observation $o_m \in \mathcal{O}_m$ is structured into three layers:

$$o_m(t) = [o_m^p(t), o_m^g(t), o_m^h(t)], \quad (8)$$

- **Private Information Layer.** $o_m^p(t) = \{s_m^d(t), u_m^p(t), a_m^h(t)\}$ captures local context for FO generation, including device states (e.g., battery SOC, EV connection status, heat pump temperature), user preferences (e.g., desired charging completion time, comfort temperature range), and historical actions that inform temporal consistency. This layer ensures agents have complete awareness of their local environment.
- **Public Information Layer.** $o_m^g(t) = \{t, m_p(t), w(t)\}$ provides system-wide context including time features (hour of day, day of week), market prices (current and predicted), and weather conditions (temperature, solar irradiance). By incorporating market price signals, agents can anticipate trading dynamics and generate FOs that align with favorable market conditions—for example, shifting flexible loads to low-price periods.
- **Cross-manager Information Layer.** $o_m^h(t) = s_{-m}^a$, where s_{-m}^a denotes the aggregated state information from all managers other than m , enables coordination awareness by providing aggregated observations from other managers. This layer allows agents to anticipate competitive and cooperative trading behavior, avoiding scenarios where multiple managers simultaneously flood the market with similar offers.

By incorporating market signals and inter-manager states, this structure enables agents to generate FOs that are informed by downstream pipeline outcomes.

Closed-loop Reward Design. To enable end-to-end optimization across the FO life-cycle, we design a multi-objective reward function that integrates stakeholder interests and propagates downstream outcomes back to the generation stage. The learning objective is defined as:

$$\max_{\{\pi_m\}} \mathbb{E}_{\tau \sim \rho^\pi} \left[\sum_{t=0}^{23} \gamma^t r_m(t) \right], \quad (9)$$

where π_m represents the policy of manager m ; τ represents trajectory samples under joint policy $\pi = (\pi_1, \dots, \pi_M)$; ρ^π is the trajectory distribution induced by the joint policy; and $r_m(t)$ is the system reward function that balances four critical objectives at time t . It is defined as:

$$r_m(t) = \alpha r_e(t) + \beta r_u(t) + \delta r_c(t) + \lambda r_o(t), \quad (10)$$

where $\alpha, \beta, \delta, \lambda \in [0, 1]$ are empirically validated objective weighting coefficients that balance economic efficiency with user satisfaction requirements; $r_e(t)$ represents the reward function for economic efficiency; $r_u(t)$ represents the reward function for user

satisfaction; $r_c(t)$ represents the reward function for constraint satisfaction; and $r_o(t)$ represents the reward function for multi-agent coordination. Their detailed definitions are as follows:

- $r_e(t)$ captures the direct monetary impact of energy consumption and generation decisions. It is defined as:

$$r_e(t) = - \sum_{d \in \mathcal{D}} \left(C_d^e(a_t^d, m_p(t)) + C_d^\eta(a_t^d, s_t^d) \right), \quad (11)$$

where $C_d^e(a_t^d, m_p(t))$ represents the energy cost for device d calculated as the product of device action a_t^d (energy consumption/generation in kWh) and market price $m_p(t)$; $C_d^\eta(a_t^d, s_t^d)$ represents efficiency losses specific to each device, determined by device action a_t^d and current device state s_t^d (e.g., battery charging efficiency degradation, heat pump coefficient of performance (COP) variations with outdoor temperature), and the negative sign transforms costs into rewards for economic optimization.

- $r_u(t)$ aggregates individual user comfort and preference satisfaction across all users $\mathcal{U}$. It is defined as:

$$r_u(t) = \sum_{u \in \mathcal{U}} w_u \cdot u_{sa}(s_t^u, u_p), \quad (12)$$

where w_u represents the preference weight for user $u \in \mathcal{U}$ derived from aggregated user preferences; and $u_{sa}(s_t^u, u_p)$ represents the user-specific satisfaction based on device state of user s_t^u and user preferences u_p . For example, in the case of EV users, the charge completion satisfaction can be computed as $\min(EV_c^{soc}/EV_g^{soc}, 1.0)$, where EV_c^{soc} and EV_g^{soc} denote the current and target state of charge (SOC), respectively.

- $r_c(t)$ ensures device operational bounds and safety requirements by penalizing actions that exceed device capabilities or violate predefined safety constraints. It is defined as:

$$r_c(t) = - \sum_{d \in \mathcal{D}} \max(0, |a_t^d| - a_{lim,d}) - \sum_{d \in \mathcal{D}} C_s^d, \quad (13)$$

where $a_{lim,d}$ denotes the maximum allowable action magnitude for device d (e.g., maximum charging power for batteries, maximum heating power for heat pumps); and C_s^d represents device-specific safety constraint violations such as SOC bounds for batteries ($b_{soc} \in [0.1, 0.9]$), temperature limits for heat pumps ($T_h \in [18^\circ\text{C}, 26^\circ\text{C}]$), and connection requirements for electric vehicles.

- $r_o(t)$ incentivizes effective inter-manager cooperation through successful market trades, bidding efficiency, and information sharing quality under Dec-POMDP communication constraints. It is defined as:

$$r_o(t) = \sum_{m \in \mathcal{M}} \left(s_m^p(t) + \eta_m^k(t) + q_m^f(t) \right), \quad (14)$$

where $s_m^p(t)$ represents the binary success indicator for manager m 's market participation (1 if successful trade, 0 otherwise); $\eta_m^k(t)$ represents the economic efficiency of manager m 's bidding strategy relative to market prices; $q_m^f(t)$ represents the quality of inter-agent communication. This reward signal propagates downstream outcomes back to the generation stage: agents generating FOs leading to successful market trades receive positive reinforcement, while those resulting in failed aggregation or trading are penalized. Combined with the cross-stage observation

Algorithm 1 Dec-POMDP Environment Execution

Input: Device control signals $\{C_d(t)\}$, current state $s(t)$
Output: Next observations $\{o_m(t+1)\}$, rewards $\{r_m(t)\}$

```

1: for episode  $n = 1$  to  $N$  do
2:   Reset environment
3:   for  $t \in T$  do
4:     for each device  $d \in \bigcup \mathcal{D}_m$  do
5:        $s_d(t+1), r_d(t) \leftarrow (s_d(t), C_d(t))$ 
6:     end for
7:     Update  $p_{t+1}, w_{t+1}, u_{sa}(s_{t+1}^d, u_p)$ 
8:     for each manager  $m \in \mathcal{M}$  do
9:        $o_m^p(t+1) \leftarrow \{s_d(t+1)\} \cup w_u$ 
10:       $o_m^g(t+1) \leftarrow (p_{t+1}, w_{t+1}, t)$ 
11:       $o_m^h(t+1) \leftarrow (\{s_{d'}(t+1) \mid d' \notin \mathcal{D}_m\})$ 
12:       $o_m(t+1) \leftarrow \text{Eq. (8)}$ 
13:    end for
14:     $r_m(t) \leftarrow \text{Eq. (10)}$ 
15:  end for
16: end for
17: return  $\{o_m(t+1)\}_{m \in \mathcal{M}}, \{r_m(t)\}_{m \in \mathcal{M}}$ 

```

structure, this mechanism closes the feedback loop and enables end-to-end optimization across the FO life-cycle.

Environment Execution with Cross-stage Feedback. Algorithm 1 illustrates how FOgym executes the Dec-POMDP environment to realize cross-stage feedback. At the beginning of each episode, the environment resets all device states, market conditions, and manager observations. For each timestep t , the environment first updates device states based on control signals $C_d(t)$ from the scheduling module, then constructs the three-layer observation for each manager as defined in Eq. (8).

The reward $r_m(t)$ is computed using Eq. (10), integrating economic efficiency, user satisfaction, constraint compliance, and coordination outcomes. Critically, this reward incorporates feedback from downstream stages—trading success and market clearing results directly influence the reward signal, enabling generation agents to learn strategies that optimize end-to-end performance.

Additionally, the training loop of MARL, including policy updates and experience collection, is provided in Algorithm 2 (see Appendix B).

5.3 Modular FO Life-cycle

FOgym implements each stage of the FO life-cycle as an independent, interchangeable module. This modular design is motivated by two considerations: (i) different deployment scenarios may require different algorithm combinations—for example, a system prioritizing user satisfaction may prefer proportional disaggregation, while one focusing on market efficiency may choose different trading strategies; (ii) as a benchmarking platform, FOgym needs to support systematic comparison of algorithms at each stage while keeping other stages fixed. By exposing standardized interfaces, researchers can independently replace or customize algorithms at any stage without affecting others, facilitating both practical deployment and algorithm development.

Module Interface Design. To ensure seamless integration and interchangeability, FOgym defines standardized interfaces for each module. The data flow follows a sequential pipeline: Generation $\rightarrow$ Aggregation $\rightarrow$ Trading $\rightarrow$ Scheduling, with each module exposing consistent input/output specifications.

- *Generation Interface:* Input—observations $o_m(t)$; Output—device-level FOs $\{F_d(t)\}$ following Definition 1. The interface accepts any MARL algorithm that maps observations to FO parameters.
- *Aggregation Interface:* Input—individual FOs $\{F_d(t) \mid d \in \mathcal{D}_m\}$; Output—aggregated FO $F_m^A(t)$ following Definition 2. The interface supports pluggable aggregation strategies with configurable optimization objectives.
- *Trading Interface:* Input—aggregated FOs $\{F_m^A(t)\}$ from all managers; Output—successful trades $\mathcal{T}(t)$ following Definition 4. The interface accommodates various market mechanisms and bidding strategies.
- *Scheduling Interface:* Input—trade outcomes $\mathcal{T}(t)$; Output—device control signals $\{C_d(t)\}$ following Definition 5. The interface enables different disaggregation policies based on fairness or efficiency criteria.

This sequential pipeline forms a closed loop: the scheduling module's output (device control signals) affects device states, which in turn influence the next timestep's FO generation. Furthermore, the cross-stage feedback mechanism described in Section 5.2 propagates downstream outcomes (trading success, user satisfaction) back to the generation stage through the reward signal, enabling end-to-end optimization despite the modular architecture.

Aggregation Module. This module combines individual FOs from multiple devices under a manager into aggregated FOs while preserving temporal constraints and maximizing usable flexibility. It supports two strategies following Valsomatzis et al. [32]:

- The *Longest Profile (LP)* strategy prioritizes aggregated FOs with maximal duration and energy content, suitable for scenarios where energy quantity is the primary concern.
- The *Dynamic Profile (DP)* strategy maximizes time flexibility by dynamically adjusting time slices, enabling better adaptation to volatile market conditions.

Trading Module. This module enables market participation through both buying and selling bids derived from aggregated FOs. It integrates two algorithms following Valsomatzis et al. [32]:

- The *Bidding* algorithm enables market participants to express their energy buying/selling intentions and conditions, constructing competitive bids based on aggregated flexibility and market forecasts.
- The *Market Clearing* algorithm finds market-clearing prices and quantities through supply-demand balance principles, matching buyers and sellers to maximize social welfare.

Scheduling Module. This module disaggregates market-level trade outcomes into device-level control actions. It provides two strategies following Šikšnys et al. [43]:

- The *Average* strategy distributes total energy equally among all participants, ensuring uniform allocation regardless of individual contributions.

- The *Proportional* strategy distributes energy based on each user's demand weight proportionally, rewarding users who contribute more flexibility to the aggregated offer.

Details of the methods in aggregation, trading, and scheduling modules for integration are provided in Appendices B.2 to B.4.

Evaluation Metrics. To comprehensively assess the performance of FOgym across different algorithm configurations, we define four evaluation metrics corresponding to different stakeholders and pipeline stages: user satisfaction measures end-user experience, economic benefit captures financial outcomes, aggregation score evaluates the quality of FO combination and disaggregation, and trading score assesses market participation effectiveness. Together, these metrics provide a holistic view of the FO life-cycle performance.

- User Satisfaction (g_u). In home energy management, ensuring that user energy demands are adequately met is a primary objective. We quantify user satisfaction as the ratio between actual energy allocation and user demand:

$$g_u = \min \left(1, \frac{E_a}{E_d} \right), \quad (15)$$

where E_a represents the total energy allocated to the user through the scheduling process, and E_d represents the user's requested energy demand. The min function ensures that over-allocation does not artificially inflate satisfaction scores. A value approaching 1.0 indicates that user demands are fully satisfied, while lower values suggest unmet energy requirements.

- Economic Benefit (g_e). Beyond satisfying energy demands, users expect economic benefits from participating in flexible energy management. We measure the monetary savings achieved through optimized energy trading and scheduling:

$$g_e = (R_t - C_t) \times \lambda_c, \quad (16)$$

where R_t denotes the total revenue obtained from energy trading activities, C_t represents the total operational costs including energy procurement and device operation, and λ_c is a currency conversion coefficient. In our experiments, g_e is expressed in Danish Krone (DKK), reflecting the real-world deployment context in Denmark.

- Aggregation & Disaggregation Score (g_a). The quality of FO aggregation and disaggregation significantly impacts market participation efficiency and scheduling accuracy. We design a composite metric that captures multiple dimensions of aggregation performance:

$$g_a = \omega_1 \Gamma_c + \omega_2 \left(1 - \frac{\Gamma_r}{\Gamma_r^{max}} \right) + \omega_3 (1 - \Gamma_v) + \omega_4 \Gamma_p, \quad (17)$$

where Γ_c denotes the compression ratio measuring the reduction in the number of FOs after aggregation; Γ_r is the root mean square error (RMSE) quantifying the deviation between aggregated power profiles and the target threshold; Γ_v is the coefficient of variation capturing power profile stability; Γ_p is the energy preservation ratio; $\omega_1, \omega_2, \omega_3, \omega_4$ are weighting coefficients that balance these components based on system priorities. Detailed definitions are provided in Appendix C.2.

- Trading Score (g_t). Effective market participation is essential for realizing the economic potential of aggregated flexibility. We

evaluate trading performance through:

$$g_t = \mu_1 \Gamma_s + \mu_2 \Gamma_m, \quad (18)$$

where $\Gamma_s = N_s/N_a$ represents the trade success rate with N_s being the number of successful trades and N_a being the total number of aggregated FOs; Γ_m denotes the market clearing efficiency; μ_1 and μ_2 are weighting coefficients. Additional trading metrics are detailed in Appendix C.2.

6 Experiments

6.1 Experimental Setup and Baselines

We construct the simulation environment based on the real-world distribution network topology of Gistrup, Denmark, using supporting data from Energinet¹, as illustrated in Fig. 5. The system coordinates 118 heterogeneous devices across 36 users, spanning 5 distinct types: Home Battery (24 units), Heat Pump (36 units), Electric Vehicle (14 units), Dishwasher (36 units), and Photovoltaic System (8 units). These devices are organized under 4 distributed managers: Manager 1 (6 users), Manager 2 (10 users), Manager 3 (8 users), and Manager 4 (12 users). Each time step $t \in \{0, 1, \dots, T-1\}$ represents one hour with $T = 24$ hours per day.

The FOgym environment is fully configurable: users can customize the number of managers, devices per manager, device types, and user preferences to construct diverse testing scenarios. The configuration described above represents one representative setup for our experiments. Detailed device parameters and training hyperparameters are provided in Appendix C.1 and C.2, respectively.

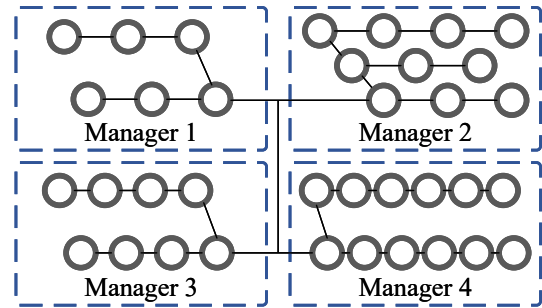


Figure 5: Distribution network topology with 36 users under 4 managers.

We integrate five MARL algorithms into the FOgym framework to support model-free FO generation, covering different coordination paradigms: on-policy methods with shared policy (MAPPO [36]) and independent policies (MAIPPO [3]), off-policy methods with centralized critics (MADDPG [17], MATD3 [1]), and fairness-aware coordination (SQDDPG [33]). This selection enables systematic comparison across different MARL design choices. We compare their performance against a model-based FO generation method [15] that relies on explicit device modeling, serving as the baseline for evaluating the benefits of model-free approaches.

¹<https://en.energinet.dk/>

Method	Model-based								MAPPO							
	LMA	LMP	LBA	LBP	DMA	DMP	DBA	DBP	LMA	LMP	LBA	LBP	DMA	DMP	DBA	DBP
A & DisA	240.07	238.69	246.18	243.67	241.22	243.08	239.79	246.53	397.04	385.67	410.51	395.33	380.42	369.69	377.04	382.25
Trading	113.69	113.34	119.79	119.68	113.87	118.45	113.39	119.56	193.80	179.15	193.01	184.88	200.36	201.58	203.89	207.22
Eco	8.38	8.64	8.47	8.56	9.32	9.01	9.11	9.24	11.07	11.39	11.17	11.22	12.25	11.83	12.05	12.14
User	0.81	0.80	0.84	0.84	0.81	0.83	0.83	0.84	0.93	0.90	0.93	0.93	0.94	0.94	0.94	0.94
Reward	533.87	532.26	577.24	578.04	533.97	572.78	572.47	573.48	726.49	723.41	724.28	723.67	727.38	727.51	727.73	727.94

Method	MADDPG								MAIPPO							
	LMA	LMP	LBA	LBP	DMA	DMP	DBA	DBP	LMA	LMP	LBA	LBP	DMA	DMP	DBA	DBP
A & DisA	401.91	411.87	398.62	403.5	363.91	392.00	387.24	399.54	403.50	401.25	367.75	407.53	384.25	409.00	372.33	402.91
Trading	187.21	199.11	190.46	192.33	195.46	209.52	212.14	219.42	190.08	188.63	174.27	190.76	204.81	214.04	204.29	220.06
Eco	10.78	11.08	10.85	10.92	11.92	11.52	11.69	11.78	10.35	10.64	10.44	10.53	11.43	11.05	11.22	11.31
User	0.93	0.94	0.93	0.93	0.94	0.94	0.94	0.95	0.93	0.92	0.91	0.93	0.93	0.95	0.93	0.96
Reward	724.33	740.61	736.14	737.72	737.48	733.19	734.04	737.18	728.84	728.39	726.47	727.23	728.49	729.41	727.91	728.62

Method	MATD3								SQDDPG							
	LMA	LMP	LBA	LBP	DMA	DMP	DBA	DBP	LMA	LMP	LBA	LBP	DMA	DMP	DBA	DBP
A & DisA	346.75	380.75	374.09	372.63	370.02	406.04	409.58	394.55	408.42	364.03	392.91	375.67	384.87	385.58	391.04	383.71
Trading	160.96	179.17	173.01	181.52	205.57	217.49	220.44	219.74	200.38	172.21	177.87	174.54	208.45	199.44	201.58	207.78
Eco	10.79	11.09	10.83	10.92	12.04	11.64	11.83	11.92	11.10	11.42	11.19	11.27	12.37	11.96	12.16	12.25
User	0.88	0.90	0.90	0.91	0.94	0.95	0.95	0.95	0.94	0.90	0.90	0.90	0.94	0.94	0.94	0.94
Reward	730.58	732.44	732.09	733.71	735.34	736.42	736.86	736.53	743.05	735.92	737.18	736.37	748.22	743.69	744.31	747.88

Table 1: Performance metrics of different method combinations in FO life-cycle. The first letter indicates the FO aggregation strategy, with L, D representing LP and DP strategies. The second letter indicates the FO trading method, with M, B representing market clearing and bidding methods. The third letter indicates the FO disaggregation strategy, with A, P representing average and proportional strategies. A & DisA (g_a , Eq. 17) and Trading (g_t , Eq. 18) represent the aggregation & disaggregation score and trading score, respectively; Eco (g_e , Eq. 16) represents the monetary savings in DKK achieved by users; User (g_u , Eq. 15) represents the user satisfaction.

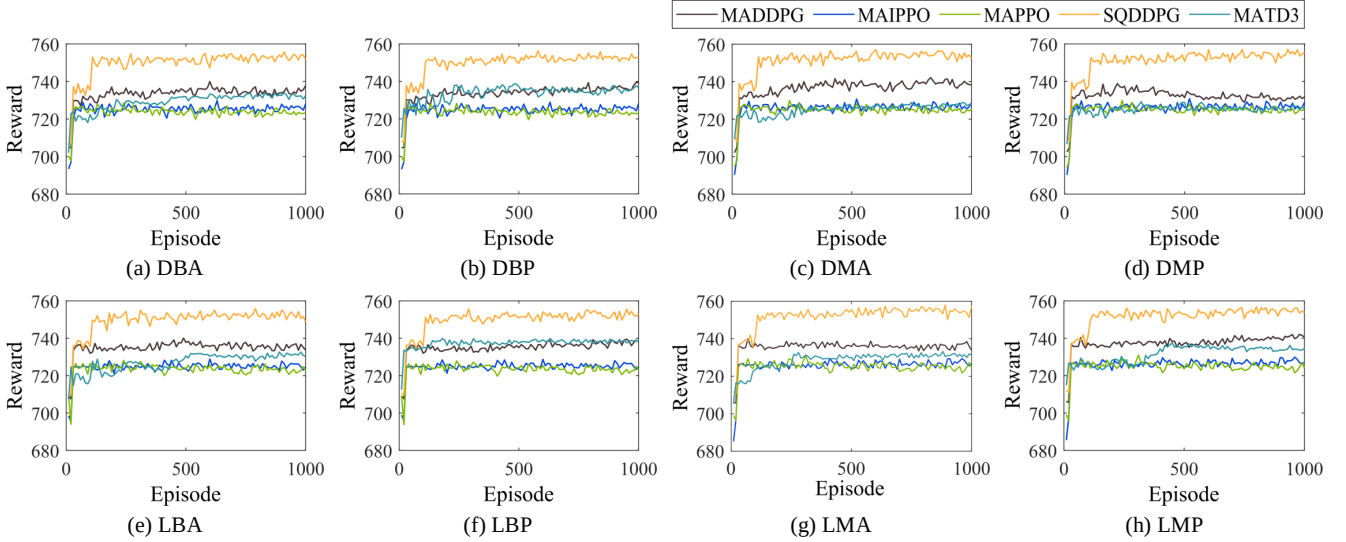


Figure 6: Performance of MARL algorithm training rewards with different combinations of algorithms in FOgym.

6.2 Main Results

Performance Analysis. We conducted comprehensive testing of the full FOgym life-cycle pipeline by evaluating five FO generation methods, two aggregation methods, two trading strategies, and two disaggregation methods. These components were flexibly combined within the modular FO pipeline framework to enable systematic benchmarking across configurations. The experimental setups and corresponding results are presented in Table 1.

As shown in Table 1, the experimental results suggest that MARL-based methods significantly outperform the model-based baseline. We now provide a concise algorithm-by-algorithm summary of the main findings:

- **MAPPO** demonstrates stable performance with low training variance (3.28). Compared to the model-based method, it improves score and total reward by 42.34% and 22.97%, yielding 2.8 DKK (31.67%) user cost savings. The shared-policy architecture and trust-region constraints effectively suppress overly aggressive

updates, enhancing policy consistency during long-horizon training.

- **MAIPPO** achieves the lowest training variance (0.77) among all methods. It improves score and total reward by 39.22% and 23.20%, yielding 2.03 DKK (22.96%) user cost savings. By assigning independent policy networks to each agent, MAIPPO enables personalized learning and reduces cross-agent interference, leading to better total rewards and user satisfaction compared to MAPPO.
- **MADDPG** outperforms the model-based method by 39.74% in score and 23.91% in total reward, offering 2.48 DKK (28.05%) user cost savings. Its deterministic policy structure and variance-reducing critic significantly improve sample efficiency. However, its training variance (21.13) is notably higher than on-policy methods.
- **MATD3** improves score and total reward by 37.75% and 23.83%, resulting in 2.54 DKK (28.73%) user cost savings. Although slightly underperforming MADDPG in total reward, it achieves stronger robustness and training stability (4.27× lower variance than MADDPG) through twin Q-networks and delayed policy updates, which mitigate Q-value overestimation bias.
- **SQDDPG** achieves the best overall performance, improving score and total reward by 37.97% and 24.64% over model-based methods, with 2.88 DKK (32.58%) user cost savings. It leverages Shapley-value-based credit assignment from cooperative game theory, embedding fairness directly into the Q-function. This promotes equitable credit distribution and stronger collaboration among agents, aligning well with the cooperative multi-agent coordination paradigm in FOGym. Its training variance (21.80) is similar to MADDPG, indicating that both algorithms have limited training stability.

Based on these results, we provide the following guidelines for algorithm selection in different deployment scenarios:

- **Training stability and consistent performance:** MAPPO and MAIPPO are recommended due to their low variance and robust convergence. MAPPO is particularly suitable for homogeneous manager configurations, while MAIPPO excels when managers control diverse device portfolios.
- **Limited training samples or fast adaptation:** MADDPG offers superior sample efficiency through off-policy learning and experience replay, enabling rapid policy improvement with fewer environment interactions.
- **Noisy observations or volatile market conditions:** MATD3 provides enhanced robustness through its twin Q-network architecture, making it suitable for real-world deployments where sensor noise and market uncertainty are prevalent.
- **Fairness and cooperative behavior among multiple stakeholders:** SQDDPG is the preferred choice. Its Shapley-value-based credit assignment ensures equitable reward distribution, promoting balanced participation in energy trading markets.
- **Large-scale deployments:** On-policy methods (MAPPO, MAIPPO) offer better scalability due to their lower memory requirements, while off-policy methods (MADDPG, MATD3, SQDDPG) may face challenges with replay buffer size as the number of agents increases.

Training Trajectory Analysis. Fig. 6 illustrates the training trajectories of various MARL algorithms under different FOGym

configurations (Fig. 6 (a) to (h)). SQDDPG consistently achieves the best performance, demonstrating higher total reward compared to all other algorithms. MAPPO and MAIPPO exhibit similar training trends, with MAIPPO occasionally surpassing MAPPO due to its independent policy structure. MADDPG and MATD3 deliver intermediate performance, falling between SQDDPG and the MAPPO variants.

These differences are due to the following factors:

- SQDDPG integrates a Shapley-value-based credit assignment mechanism, enhancing cooperation among agents and accelerating convergence through fair reward distribution.
- MAPPO achieves strong stability by employing shared policy networks and trust-region constraints, making it ideal for long-horizon tasks, albeit with limited flexibility.
- MAIPPO improves adaptability through independent policy networks, enabling personalized learning for better performance.
- MADDPG, as an off-policy algorithm, leverages deterministic policy gradients and variance-reducing critics to achieve high sample efficiency and precise control.
- MATD3 enhances robustness in noisy environments through the use of twin Q-networks and delayed policy updates, effectively mitigating overestimation bias.

All algorithms eventually converge to a stable high-reward region, indicating successful training across configurations. Model-free FO generation methods based on MARL consistently outperform the selected model-based baseline, confirming the effectiveness of our MARL-based method. These results validate FOGym as a home energy management framework and flexible open-source environment for evaluating diverse MARL algorithms across various stages of the FO life-cycle.

The performance variation across different module combinations further validates our unified pipeline design. Agents integrated with the full FO life-cycle, receiving downstream signals through observations and rewards, achieve 22.97%-24.64% improvement in total reward over model-based methods. This demonstrates that integrating MARL uniformly with the FO pipeline through cross-stage feedback enables effective end-to-end optimization. A detailed case study demonstrating the performance of SQDDPG combined with DMA algorithms is provided in Appendix A.2.

7 Conclusion

In this paper, we propose FOGym, a unified bottom-up home energy management framework integrating MARL within the FO life-cycle. FOGym employs a novel model-free FO generation approach, eliminating reliance on explicit modeling and enabling real-time adaptive decision-making through direct environment interaction. Additionally, FOGym integrates MARL uniformly with the FO pipeline through cross-stage observation spaces and closed-loop reward design, enabling generation agents to perceive downstream outcomes and achieve end-to-end optimization. Extensive experiments demonstrate that MARL-based methods achieve up to 24.64% improvement in total reward and 32.58% user cost savings compared to model-based approaches, while different algorithms exhibit distinct strengths suited to various deployment scenarios. FOGym establishes a scalable, reproducible platform for future research in adaptive bottom-up energy management.

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A Examples

A.1 An Example in FO life-cycle

The following example illustrates how an EV participates in the FO life-cycle, from FO generation to final scheduling.

Example 1. Consider an EV user who returns home at 18:00 and connects their vehicle to a home charging station. The EV has a 60 kWh battery with 20% state-of-charge (SOC) upon arrival, and the user requires at least 80% SOC by 7:00 the next morning for their commute, defining a 13-hour charging window with a minimum energy requirement of 36 kWh. At 18:00, the system generates an FO for this EV: $F_{EV}(18) = ([18 : 00, 7 : 00], [36, 48] \text{ kWh}, 0.85)$, where 36 kWh represents the minimum charge needed to reach 80% SOC, 48 kWh represents the maximum possible charge to reach full capacity, and 0.85 is the priority weight factor indicating the user's preference for fulfilling this charging request relative to other flexible loads. A higher weight means the system prioritizes satisfying this demand during scheduling and disaggregation. This FO is updated hourly as the EV charges and conditions change. The regional manager then collects FOs from multiple devices under its management and aggregates them into a combined FO representing the total energy demand. During trading, the manager submits a bid to purchase energy at a competitive price. Meanwhile, another manager with surplus energy from solar panels or battery storage submits a sell offer. When the buy and sell bids match, a bilateral trade is completed. After successful trading, the purchased energy is disaggregated back to individual devices. The EV receives a control signal instructing it to charge at 7.2 kW between 22:00 and 03:00 when electricity prices are lowest. By 7:00, the EV reaches 85% SOC, satisfying the user's requirement while minimizing energy costs through participation in the flexibility market.

A.2 An Example Case

Take the case of the SQDDPG combined with DMA algorithms in the FOgym. The experiment involves 36 users coordinated by 4 regional manager agents, operating within a 24-hour FO life-cycle. After 1000 episodes training, over the 24-hour operational cycle, the system executed a total of 312 energy transactions, averaging 13 trades per hour. Transaction peaks occurred at hours 4, 7, 11, 13, 14, 17, and 20, with up to 26 trades per peak hour, while the lowest activity was observed at hour 3 with only 6 trades. The cumulative transaction value reached 4,452.89 (DKK), with an average of 14.27 (DKK) per trade. The system successfully generated and scheduled 9,039 FOs, enabling fine-grained demand-side energy management. User satisfaction was consistently maintained more than 90 % across all hours, demonstrating that the system reliably fulfilled user energy requirements and comfort preferences. Economically, the system yielded significant user cost savings, each user saved an average of 12.37 (DKK) compared to flat-rate pricing schemes. On the supply side, the improved allocation of flexibility resources led to an estimated 12 % reduction in peak generation costs. Additionally, based on the regional carbon intensity of 0.5-0.75kg CO₂/kWh, the system is estimated to have reduced carbon emissions by 445-670kg CO₂. These results validate the practical viability of SQDDPG combined with DMA algorithms in FOgym for home energy management.

Algorithm 2 Multi-Agent FO life-cycle

Input: Manager observations $\{o_m(t)\}_{m \in \mathcal{M}}$, device portfolios $\{\mathcal{D}_m\}$, Number of episodes N , Time step T , learning rates $\{\alpha_m\}$, exploration parameters $\{\epsilon_m\}$

Output: Device control $\{C_d(t)\}_{d \in \cup \mathcal{D}_m}$, Updated policies $\{\pi'_m\}_{m \in \mathcal{M}}$

```

1: for episode  $n = 1$  to  $N$  do
2:   Reset environment, get  $\{o_m(0)\}_{m \in \mathcal{M}}$ 
3:   for  $t \in T$  do
4:     for each manager  $m \in \mathcal{M}$  do
5:       for each device  $d \in \mathcal{D}_m$  do
6:         Generate  $F_d(t) \leftarrow a_m(t) \leftarrow \pi_m(o_m(t))$ 
7:       end for
8:       Compute  $F_m^A(t) \leftarrow F_d(t)$ 
9:       Generate  $B_m(t) \leftarrow F_m^A(t), p_m^b(t), k_m(t)$ 
10:       $\mathcal{T}(t) \leftarrow M_c(B_m(t))$ 
11:      if  $T_m \in \mathcal{T}(t)$  then
12:        Generate  $C_d(t) \leftarrow (P_d^c(t), t_d^s, t_d^e)$ 
13:      else
14:         $C_d(t) \leftarrow$  Energy grid
15:      end if
16:       $s_{t+1}, r_m^t \leftarrow \{C_d(t)\}$ 
17:      Store  $\{o_m^t, a_m^t, r_m^t, o_m^{t+1}\}$  in ReplayBuffer
18:      if update conditions met then
19:         $\{\pi'_m\} \leftarrow \{\pi_m\}$ 
20:      end if
21:    end for
22:  end for
23:   $\epsilon'_m \leftarrow \epsilon_m, \sigma'_e \leftarrow \sigma_e$ 
24: end for
25: return  $\{C_d(t)\}_{d \in \cup \mathcal{D}_m}, \{\pi'_m\}_{m \in \mathcal{M}}$ 

```

B Details of FOgym Algorithm Integration

FOgym adopts a standardized Dec-POMDP formulation that unifies observation spaces, action spaces, and reward structures, enabling fair and consistent comparison across diverse MARL paradigms. Based on this formulation, FOgym integrates MARL agents within the full FO life-cycle, as illustrated in Algorithm 2.

B.1 MARL Methods for Integration

MAPPO. To accommodate the structural characteristics of the FO environment, we implement trust region optimization with FO-specific constraint-aware modifications. The architecture employs separate actor and critic networks with shared convolutional layers for encoding device state sequences. The actor network $\pi_\theta(a|o)$ outputs device-type-specific action parameters via modular heads, while the critic network $V_\phi(o)$ incorporates constraint satisfaction into the value estimation.

The FO-compliant MAPPO variant, specifically tailored for FO generation, is defined as:

$$\mathcal{L}^\pi = \mathcal{L}_p + \lambda_v \cdot \mathcal{L}_v - \lambda_e \cdot \mathcal{H} + \lambda_f \cdot \mathcal{L}_f + \lambda_t \cdot \mathcal{L}_t, \quad (19)$$

where $\mathcal{L}_p$ represents the PPO clipped objective function is defined as:

$$\mathcal{L}_p = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (20)$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ represents the probability ratio between new and old policies; $\hat{A}_t = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}$ represents the Generalized Advantage Estimation (GAE); $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ represents the TD error; $\mathcal{L}_v = \frac{1}{2} (V_\theta(s_t) - V_t^T)^2$ represents the value function loss; $\mathcal{H} = -\sum_a \pi_\theta(a|s_t) \log \pi_\theta(a|s_t)$ represents the policy entropy; $\mathcal{L}_f$ represents the FO constraint loss; $\mathcal{L}_t$ represents the market constraint loss.

MAPPO exhibits high training stability due to its trust region constraint, which effectively limits excessive policy updates. The use of a shared policy network improves parameter efficiency and reduces variance while naturally promoting coordinated behavior among agents. These properties make it particularly well-suited for long-horizon planning tasks, such as 24-hour operational cycles, where robust convergence and stable training are critical.

MAIPPO. To address agent-specific heterogeneity in FO environments, we implement an independent policy optimization scheme where each manager maintains its own actor-critic networks and local replay buffer. The actor $\pi_{\theta_m}(a|o)$ performs device-level control under local partial observations, while the critic $V_{\phi_m}(o)$ estimates value functions independently without parameter sharing or cross-agent coupling. By fully decoupling policy updates, this design mitigates inter-agent interference and enables personalized policy adaptation to user-device diversity.

The FO-adapted MAIPPO variant, designed to evaluate agent-level strategy specialization and conflict resolution within the FO pipeline, is defined as:

$$\{\pi_m^*\}_{m=1}^M = \{\arg \max_{\pi_m} \mathbb{E}[\mathcal{L}_p(\pi_m) + \lambda_f^m \mathcal{L}_f^m]\}_{m=1}^M, \quad (21)$$

where each Manager m independently optimizes its own policy π_m . The loss function for each policy is defined as:

$$\mathcal{L}_p(\pi_m) = \mathbb{E}_t [\min(r_t^m(\theta_m) \hat{A}_t^m, \text{clip}(r_t^m(\theta_m), 1 - \epsilon, 1 + \epsilon) \hat{A}_t^m) + \lambda_v^m \mathcal{L}_v^m - \lambda_e^m \mathcal{H}^m]$$

And the adaptive learning rate adjustment is defined as:

$$\alpha_m^{t+1} = \max(\alpha_m^t \cdot I_d, \alpha_{\min}) \quad (22)$$

MAIPPO mitigates policy interference by assigning independent policy networks to each manager, enabling the learning of personalized strategies. It offers strong adaptability to heterogeneous user and device configurations, making it well-suited for scenarios with highly differentiated coordination tasks across agents.

MADDPG. We implement deterministic policy gradients augmented with FO-specific critic regularization and sample-efficient replay. The actor network $\mu_\theta(o)$ generates deterministic control actions under device-specific bounds, incorporating batch normalization to stabilize learning. The critic network $Q_\phi(o, a)$ leverages temporal difference updates, further enhanced with constraint regularization to penalize violations of the FO parameter space, defined as $L_c = \lambda \sum_d \max(0, |a_d| - 1)^2$.

The FO-tailored MADDPG variant, designed for constraint-aware optimization in continuous action spaces, is defined as:

$$\pi_F^D = \arg \max_{\pi} \mathbb{E}_{s,a \sim F} [Q^F(s, a)], \quad (23)$$

where $Q^F(s, a)$ represents the state-action value function.

The actor network is updated using the deterministic policy gradient:

$$\nabla_{\theta_\pi} J(\pi) = \mathbb{E}_{s \sim \mathcal{D}} [\nabla_a Q(s, a)|_{a=\pi(s)} \nabla_{\theta_\pi} \pi(s)] \quad (24)$$

The critic network is updated using temporal difference (TD) learning:

$$\mathcal{L}(\theta_Q) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} [(Q(s, a) - y)^2], \quad (25)$$

where $y = r + \gamma Q'(s', \pi'(s'))$ is the TD target.

The target network soft update is defined as:

$$\theta' \leftarrow \tau \theta + (1 - \tau) \theta'. \quad (26)$$

MADDPG achieves high sample efficiency by leveraging off-policy learning and historical experience, while deterministic policies enable precise control in continuous action spaces. Value-based critics reduce variance, leading to stable returns. It is particularly effective in sample-constrained environments that require rapid policy adaptation and continuous control optimization.

MATD3. To meet the stability demands of energy market control, we implement a twin-delayed policy gradient algorithm tailored for FO constraints. MATD3 adopts a double Q-network architecture— $Q_{\phi_1}(o, a)$ and $Q_{\phi_2}(o, a)$ —to mitigate overestimation bias, and incorporates delayed actor updates to promote stable convergence. Target policy smoothing is applied as $\tilde{a} = \text{clip}(\mu_{\theta'}(o') + \text{clip}(\epsilon, -c, c), -1, 1)$, with $\epsilon \sim \mathcal{N}(0, \sigma^2)$ introducing regularized exploration.

The FO-specific adaptation of MATD3, designed to ensure stable policy learning in noisy market environments, is defined as:

$$Q^T(s, a) = \min(Q_1^T(s, a), Q_2^T(s, a)), \quad (27)$$

where Q_1^T and Q_2^T represent two independent critic networks.

The target action calculation, which includes noise regularization, is defined as:

$$a' = \pi'(s') + \text{clip}(\epsilon, -c, c), \quad \epsilon \sim \mathcal{N}(0, \sigma) \quad (28)$$

The TD target calculation is defined as:

$$y = r + \gamma \min_{i=1,2} Q_i'(s', a') \quad (29)$$

The delayed policy update: Actor network and target networks are updated every d steps (typically $d = 2$).

MATD3 improves numerical stability by employing twin Q-networks to mitigate overestimation bias, and demonstrates strong robustness to observation noise and hyperparameter variations. It achieves stable convergence in complex environments, making it particularly well-suited for high-noise scenarios requiring reliable long-term performance.

SQDDPG. We integrate cooperative game theory into the FO environment by SQDDPG with Shapley value-based credit assignment for fair multi-agent coordination. SQDDPG computes each agent's marginal contribution across all possible coalitions, embedding the resulting Shapley values into the critic network's learning objective. This promotes equitable benefit distribution among collaborating manager agents. The fairness component integrated into the critic loss is defined as:

$$\mathcal{L}^S = \mathbb{E}[(Q(o, a) - (r + \gamma Q'(o', a'))^2] + \lambda_s \cdot |\phi_m - \hat{\phi}_m|^2, \quad (30)$$

Algorithm 3 MARL Policy Update

Input: Number of episodes N , learning rates $\{\alpha_m\}$, exploration parameters $\{\epsilon_m\}$
Parameter: $Generation \in \text{MARL}$

Output: Updated policies $\{\pi'_m\}_{m \in \mathcal{M}}$

- 1: Initialize $\{\pi_m\}$, buffers, and network
- 2: **if** update conditions met (every K steps or batch size requirements) **then**
- 3: **for** each manager $m \in \mathcal{M}$ **do**
- 4: **if** MAPPO **then**
- 5: $\pi_m \leftarrow \pi_m + \alpha_m \nabla_{\theta} \mathcal{L}_p(\mathcal{B})$
- 6: **else if** MAIPPO **then**
- 7: $\pi_m \leftarrow \pi_m + \alpha_m \nabla_{\theta} \mathcal{L}_p(\mathcal{B}_m)$
- 8: **else if** MADDPG **then**
- 9: Update Critic: $Q_m \leftarrow Q_m - \alpha_m \nabla_{\phi} \mathcal{L}_q(\mathcal{B}_m)$
- 10: Update Actor: $\pi_m \leftarrow \pi_m + \alpha_m \nabla_{\theta} J(\pi_m)$
- 11: **else if** MATD3 **then**
- 12: Update Twin Critics: $Q_m^1, Q_m^2 \leftarrow (\mathcal{B}_m)$
- 13: **if** update counter % 2 == 0 **then**
- 14: Update Actor: $\pi_m \leftarrow \pi_m + \alpha_m \nabla_{\theta} J(\pi_m)$
- 15: Update target networks: $\pi_m^t, Q_m^t \leftarrow \tau \pi_m + (1 - \tau) \pi_m^t, \tau Q_m + (1 - \tau) Q_m^t$
- 16: **end if**
- 17: **else if** SQDDPG **then**
- 18: $\pi_m \leftarrow \text{Shapley}(\phi_m)$
- 19: **end if**
- 20: **end for**
- 21: **end if**
- 22: **return** Updated policies $\{\pi_m\}_{m \in \mathcal{M}}$

where $\phi_m, \hat{\phi}_m$ represent the computed Shapley value and target value, ensure equitable reward distribution. The Shapley value calculation is defined as:

$$\phi_i = \sum_{S \subseteq \mathcal{M} \setminus \{i\}} \frac{|S|!(|\mathcal{M}| - |S| - 1)!}{|\mathcal{M}|!} [v(S \cup \{i\}) - v(S)], \quad (31)$$

where $v(S)$ is the value function for coalition S , representing the total reward that coalition can achieve.

SQDDPG implements fair credit assignment within the multi-stakeholder energy market context is defined as:

$$\phi_m^F = Q_{\text{Shapley}}(m, F_{\text{env}}, \mathcal{M}), \quad (32)$$

where ϕ_m^F is the Shapley value for Manager m , representing its contribution to the global reward.

The coalition value calculation is defined as:

$$v(S) = Q_{\text{joint}} \cdot \frac{\sum_{i \in S} c_i}{\sum_{i \in \mathcal{M}} c_i}, \quad (33)$$

where Q_{joint} is the joint Q-value and c_i is agent i 's contribution.

SQDDPG incorporates Shapley-value-based credit assignment to ensure fairness in multi-agent collaboration, promoting cooperative behavior through equitable reward distribution. It enhances transparency by explicitly attributing agent contributions, making it well-suited for multi-stakeholder scenarios where fairness guarantees are critical, while maintaining strong overall learning performance.

B.2 Aggregation Methods for Integration

LP (Longest Profile) Aggregation Strategy. The LP aggregation strategy constructs aggregated FOs by maximizing profile size while preserving temporal alignment. It emphasizes energy quantity by selecting time slices with significant energy potential, and evaluates aggregation success based on reductions in root mean square error (RMSE) and coefficient of variation. This method is particularly effective in scenarios where energy demand predictions are accurate and time flexibility is less constrained. While it enables high participation rates and efficient energy consolidation, it may reduce temporal flexibility due to its emphasis on profile length.

The strategy flow works as follows: (i) Find all FOs with the maximum profile size. (ii) Select the FO with highest time flexibility as the initial FO. (iii) Add all other FOs to the processing set. (iv) Iteratively perform binary aggregation, adding FOs that improve RMSE and CV.

DP (Dynamic Profile) Aggregation Strategy. The DP aggregation strategy prioritizes flexibility preservation by selecting initial offers based on a time-flexibility index and dynamically adjusting aggregation boundaries to accommodate heterogeneous device constraints. By emphasizing the retention of high-flexibility time slices, it maintains adaptability in scheduling under uncertain temporal conditions. This approach is well suited for scenarios where time availability is uncertain but energy demands are well characterized. While it offers improved temporal flexibility, it may reduce total energy capacity due to conservative aggregation boundaries.

The strategy flow works as follows: (i) Sort FOs based on time flexibility. (ii) Select the FO with the highest flexibility as the initial FO. (iii) Dynamically adjust time boundaries to maximize flexibility. (iv) Iteratively aggregate, maintaining high flexibility.

B.3 Trading Methods for Integration

Market Clearing Method. The market clearing strategy determines transaction outcomes by identifying the intersection of aggregated supply and demand curves, where the clearing price and quantity are computed to maximize economic efficiency. It enables rigorous welfare analysis through the quantification of consumer and producer surplus. This method is particularly well suited for mature energy markets with high liquidity and diverse participation, offering efficient price discovery and theoretically optimal social welfare. However, its effectiveness relies on the presence of sufficient market participants to ensure stable clearing dynamics.

The method flow works as follows: (i) Collect bids from all buyers and sellers. (ii) Construct supply and demand curves. (iii) Find the intersection point to determine clearing price. (iv) Match buyers and sellers based on clearing price. (v) Generate trades that satisfy supply-demand balance.

Bidding Method. The Bidding algorithm enables decentralized bid management by allowing market participants (managers) to formulate buy and sell offers based on local preferences and dynamic market conditions. It supports multiple bid types—including fixed, block, and curve bids—and incorporates a robust validation mechanism to ensure compliance with market rules. The framework maintains participant registration and bid metadata while

providing real-time market statistics to support informed decision-making. This design facilitates flexible and secure bidding behavior within heterogeneous, multi-agent energy markets.

The method flow works as follows: (i) Register market participants. (ii) Collect bids from participants. (iii) Validate and organize bids. (iv) Process bids and generate trades.

B.4 Disaggregation Methods for Integration

Average Disaggregation Strategy. The average disaggregation strategy offers a simple and intuitive mechanism by uniformly distributing energy across participating devices without requiring additional parameters. This method is computationally efficient and ensures fairness, making it suitable for homogeneous device groups or scenarios where equal treatment is prioritized. However, it does not account for individual device capabilities such as power ratings or storage limits, which may result in suboptimal resource utilization or energy waste. Given total energy E and number of users N , each user receive is defined as:

$$E_i = \frac{E}{N}, \quad \forall i = 1, 2, \dots, N \quad (34)$$

Proportional Disaggregation Strategy. The proportional disaggregation strategy allocates energy based on device-specific weights, such as capacity or contribution, thereby aligning resource distribution with individual capabilities. This method enhances efficiency and minimizes the risk of ineffective allocation, particularly in heterogeneous device groups. However, it requires prior estimation of device weights and introduces additional implementation complexity. Let user i 's weight be w_i and total weight $W = \sum w_j$, then the user receive is defined as:

$$E_i = \frac{w_i}{W} \times E \quad (35)$$

C Details of Experimental Settings

This appendix provides detailed experimental configurations, including device parameters and MARL algorithm hyperparameters.

C.1 Device Types and Parameter Settings

We coordinate 118 heterogeneous devices across 36 users, spanning 5 distinct types with varying constraints, temporal behaviors, and operational requirements. A unified interface abstracts this heterogeneity while preserving essential device characteristics for FO generation. The system is organized under 4 distributed managers with user allocation as follows: Manager 1 (6 users), Manager 2 (10), Manager 3 (8), and Manager 4 (12), with each user managing an average of 3.3 devices.

Home Battery Storage System (24/36 users). The battery, as a core component of the home energy system, performs peak shaving and valley filling by charging at low prices and discharging at high prices, thereby improving energy efficiency. The battery parameters are shown in Tab. 2.

Parameter	Range	Parameter	Range
SOC	[0.1, 0.9]	η	[0.8, 0.95]
p (kW)	[3.0, 7.0]	SOC_s	[0.1, 0.9]
Capacity (kWh)	[5, 15]	Type	Lithium-ion

Table 2: Battery Parameter Settings

In the Tab. 2, SOC represents State of Charge of the battery, p represents the charging/discharging power limits based on battery capacity and health status, η represents the efficiency of the battery, and SOC_s represents the initial SOC of the battery. The battery rewards structure is defined as:

$$r_b(t) = \alpha_b \times r_b^e(t) + \beta_b \times r_b^\eta(t) + \gamma_b \times r_b^l(t), \quad (36)$$

where the $\alpha_b, \beta_b, \gamma_b \in [0, 1]$ represent the battery factors, $r_e(t)$, $r_\eta(t)$, $r_l(t)$ represent the reward function of economic, efficiency, and maintenance, respectively.

Heat Pump System (36/36 users). Heat pumps, commonly installed in residential buildings, regulate indoor temperature by adjusting heating power to maintain thermal comfort. The heat pump parameters are shown in Tab. 3.

Parameter	Range	Parameter	Range
COP	[3.0, 5.0]	T_c ($^{\circ}C$)	[18.0, 26.0]
P (kW)	[2.0, 9.0]	C_h (kWh/ $^{\circ}C$)	[8.0, 15.0]
T_c^{sp} ($^{\circ}C$)	21		

Table 3: HeatPump Parameter Settings

In the Tab. 3, T_c represents the temperature in the room, C_h represents the thermal capacity of the room, indicating how much energy is required to raise the room temperature by $1^{\circ}C$, and T_c^{sp} represents the setpoint temperature specified by the users. The heat pump reward structure is defined as:

$$r_h(t) = \alpha_h \times r_h^e(t) + \beta_h \times r_h^c(t), \quad (37)$$

where the $\alpha_h, \beta_h \in [0, 1]$ represent the heat pump factors, $r_h^e(t)$, $r_h^c(t)$ represent the reward function of economic, and user comfort, respectively.

Electric Vehicle (EV) (14/36 users). The EV model includes charging-related parameters but omits discharging due to practical constraints. Energy allocation is computed within the time window between connection and departure. The EV parameters are shown in Tab. 4.

Parameter	Range	Parameter	Range
Capacity (kWh)	[20.0, 100.0]	SOC	[0.1, 0.9]
p (kW)	[3.7, 11.0]	η	[0.85, 0.95]
t_c (h)	(0, 1)	t_d (h)	(0, 1)

Table 4: EV Parameter Settings

In the Tab. 4, p represents the charging power of the EV, t_c and t_d represent the connection time and the departure time of the EV, respectively. The EV reward structure is defined as:

$$r_{ev}(t) = \alpha_{ev} \times r_{ev}^e(t) + \beta_{ev} \times r_{ev}^s(t) + \gamma_{ev} \times r_{ev}^a(t), \quad (38)$$

where the $\alpha_{ev}, \beta_{ev}, \gamma_{ev} \in [0, 1]$ represent the EV factors, $r_{ev}^e(t)$, $r_{ev}^s(t)$, $r_{ev}^a(t)$ represent the reward function of economic, charging complete status, and user satisfaction, respectively.

Dishwasher (36/36 users). The dishwasher abstractly represents appliances with fixed execution duration and energy demand (e.g., dishwashers, washing machines). Once loaded by the user, the appliance enters a ready state and, upon activation, executes the full cycle with a constant energy requirement. The dishwasher parameters are shown in Tab. 5.

Parameter	Range	Parameter	Range
E_d (kWh)	[1.0, 3.0]	t_l (h)	[0.5, 6.0]
ϵ	(1,2,...5)	Interruptible	False
t_{run} (h)	[2.0, 4.0]	p_{rated} (kW)	[0.8, 1.5]

Table 5: Dishwasher Parameter Settings

In the Tab. 5, E_d represents the energy demand for the machine, t_l represents the time delay after the machine has deployed, ϵ represents the priority of the user demands, and t_{run} and p_{rated} represent the running time period of the machine and the rated power for the machine working, respectively. The dishwasher reward structure is defined as:

$$r_w(t) = \alpha_w \times r_w^c(t) + \beta_w \times r_w^p(t) + \gamma_w \times r_w^u(t) - \theta_w \times r_w^e(t), \quad (39)$$

where the $\alpha_w, \beta_w, \gamma_w, \theta_w \in [0, 1]$ represent the dishwasher factors, $r_w^c(t), r_w^p(t), r_w^u(t), r_w^e(t)$ represent the reward function of completion, progress, user satisfaction, and economic, respectively.

Photovoltaic System (PV) (8/36 users). PV systems generate electricity based on solar irradiation, which is highly sensitive to weather conditions and inherently stochastic. As there is no controllable action over solar input, PV units operate passively, supplying power to local users or generating FOs for trading. The PV parameters are shown in Tab. 6.

Parameter	Range	Parameter	Range
p (kW)	[3.0, 10.0]	η_g	[0.15, 0.22]
D_{area} (m^2)	[15.0, 35.0]	A_{tilt} ($^\circ$)	[20.0, 40.0]
A_f	0.8	w_d	True

Table 6: PV Parameter Settings

In the Tab. 6, η_g represents the efficiency of the PV generation, D_{area} represents the area of PV panel, A_{tilt} represents the tilt angle of PV panel, A_f represents the accuracy of weather forecasting, and w_d indicates whether the PV generation depends on the weather. The PV reward structure is defined as:

$$r_v(t) = \alpha_v \times \eta_v \times r_v^g(t) \times r_v^p(t), \quad (40)$$

where the $\alpha_v \in [0, 1]$ represents the PV factors, η_v represents the generation efficiency factor, and $r_v^g(t), r_v^p(t)$ represent the reward function of power generation and market price, respectively.

C.2 MARL Hyperparameter Settings

The specific parameter settings of different MARL algorithms, including MAPPO, MAIPPO, MADDPG, MATD3, and SQDDPG, are shown in the following Tabs. 7 to 11, respectively.

Batch size	Lr Actor	Lr Critic	γ	GAE_λ
256-512	0.0003	0.0003	0.99	0.95
$clip_p$	$Epoch_{ppo}$	$batch_{min}$	v_l^c	e_c
0.2	10	4	0.5	0.01

Table 7: MAPPO Hyperparameter Settings

Batch size	Lr Actor	Lr Critic	γ	GAE_λ	$clip_p$
256-512	0.0003	0.0003	0.99	0.95	0.2
$Epoch_{ppo}$	$batch_{min}$	v_l^c	e_c	lr_d	lr_{min}
10	4	0.5	0.01-0.02	0.995	$1 \times e^{-5}$

Table 8: MAIPPO Hyperparameter Settings

Batch size	Lr Actor	Lr Critic	γ
256	0.001	0.002	0.99
τ	Buffer size	θ_n	σ_n
0.01	$1 \times e^6$	0.15	0.2

Table 9: MADDPG Hyperparameter Settings

Batch size	Lr Actor	Lr Critic	γ
256-512	0.001	0.002	0.99
τ	$policy_n$	$clip_n$	$policy_f$
0.005	0.2	0.5	2

Table 10: MATD3 Hyperparameter Settings

Batch size	Lr	γ	τ
256-512	0.001	0.99	0.01
Shapley iter	Shapley b_s	Credit weight	
100	64	0.8	

Table 11: SQDDPG Hyperparameter Settings

the complexity of market participation by consolidating multiple device-level FOs into aggregated FOs. The system evaluated aggregation & disAggregation score through the following metrics: (i) RMSE, which measures the deviation of the aggregated power profile from a predefined target threshold. A lower RMSE indicates closer alignment with the target and higher aggregation quality. (ii) Compression ratio, which quantifies the degree of reduction in the number of FOs. A higher ratio reflects better aggregation efficiency and lower market complexity. (iii) Coefficient of variation, which captures the stability of the aggregated power profile. A lower value indicates more consistent power output across time slots. (iv) Energy preservation, which evaluates the extent to which the original total energy is retained after aggregation. Ideally, this value approaches 1.0, indicating minimal energy loss.

Trading Score. In the FOGym, the evaluation of trading processes is also critical for assessing overall system performance. Trading is responsible for matching aggregated FOs in the market by managers. The trading score in the main text (Eq. 18) uses two core metrics: market clearing efficiency and trade success rate. Additional metrics are provided below for comprehensive analysis: (i) Market clearing efficiency, which is defined as the ratio between successfully matched volume and total submitted bids. This metric reflects the effectiveness of the market mechanism in achieving supply-demand balance. (ii) Trade success rate, which indicates the proportion of submitted bids that result in successful transactions. Higher values denote improved market liquidity and matching efficiency. (iii) Total social welfare, which measures the

Aggregation & DisAggregation Score. In the FOGym, the evaluation of aggregation and disaggregation processes is critical for assessing overall system performance. Aggregation reduces

total economic value generated through transactions, computed as the sum of consumer and producer surplus across all cleared bids. (iv) Price discovery accuracy, which assesses the proximity

between the actual clearing price and the theoretical equilibrium price. Higher accuracy reflects more effective price discovery.

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